Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Designing Forest Light Environments in Indoor Spaces Using Dynamic Stochastic Lighting Models			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio · Medical			
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	Bio-signal Measurement / Diagnostic Device		
Concept of Proposed Technology	○ The proposed technology is a bio-inspired stochastic indoor lighting system that emulates the temporal, spatial, and spectral characteristics of light in natural forest environments, which have been shown to influence key human physiological and psychological markers, including stress, anxiety, relaxation, and blood pressure. ○ The objective would be to demonstrate these benefits within indoor spaces such as conference rooms, offices, and mixed-reality environments, where humans spend a significant amount of their waking hours under static, non-natural lighting conditions.			

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	○ It is known that the human visual system is sensitive to subtle changes in lighting, which plays a critical role in circadian regulation, mood, and overall well-being. ○ At its core, the system models key dynamic aspects of lighting in nature as a spatiotemporal stochastic process, capturing variability found in a heterogeneous outdoor environment. These characteristics are synthesized using random walk processes combined with weighted spectral functions, allowing fine control over light parameters (intensity, color temperature, and spatial distribution) across multiple indoor light fixtures. Parameters and hyperparameters will additionally be tuned using a deep learning framework to optimize output. ○ The modeled/simulated lighting signals are then reproduced in real time using tunable LED arrays with accompanying tinyML chips controlled by embedded processors, enabling multiple lighting modes (for example, morning forest, midday canopy, etc.). Optimized sensor feedback allows the system to maintain realism while avoiding flicker discomfort. ○ Potential applications include improved wellness and productive workplace environments, cognitive performance optimization, healthcare/rehabilitation spaces, and immersive lighting for AR/VR systems.
Rationale for Joint R&D Project	○ The proposed research is both highly intradisciplinary and interdisciplinary, combining aspects of stochastic modeling, lighting engineering, embedded systems, and human factors evaluation. Such expertise is difficult to develop within a single organization, and therefore, collaboration is required. ○ A joint R&D framework would enable advanced stochastic modeling and machine learning expertise from the home research institute, practical hardware prototype & design with system integration from a collaborative partner, and coordinated human-subject evaluation with accompanying physiological data analysis.
Key Words	1. Lighting and mood 2. Shinrin-yoku (Forest bathing) 3. Stochastic light modeling 4. Tunable LED systems 5. Learning-based parameter tuning

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (6)
R&D Objectives	○ Develop and simulate a stochastic/statistical model that captures the variable and heterogeneous nature of forest lighting ○ Design a real-time controller that reproduces the stochastic/statistical properties using indoor LED arrays ○ Tune the controller using optimization approaches, including machine-learning-based parameter tuning ○ Validate the system through human physiological measurements (primarily salivary cortisol, but also including variable heart rate monitoring, EEG, and hypertension monitoring			
R&D Target Specifications	○ Accurately model and simulate an experimentally gathered forest lighting condition known to have proven physiological outcomes based on published research ○ LED controller that follows simulation faithfully and conforms to IEEE 1789 specification (no perceptible flicker) ○ Physiological measurement – salivary cortisol (≥ 30-50% over baseline is stress), variable heart rate monitoring, EEG, and hypertension monitoring			
R&D Plan	○ Phase 1 → Modeling & Simulation Development (Months 1-6) ▪ Find and/or record representative experimental natural forest data ▪ Determine a natural forest environment for stochastic model development ▪ Define statistical similarity metrics • Milestone → Validate Forest light simulation model ○ Phase 2 → Controller & Hardware Design (Months 7–12) ▪ Design and integrate tunable LED hardware ▪ Implement real-time forest stochastic signal generation • Milestone → Functional indoor lighting proof-of-concept ○ Phase 3 → Optimization (Months 13-15) ▪ Optimize stochastic lighting control patterned on natural light ▪ Apply learning-based parameter tuning • Milestone → Lighting system with optimized parameters ○ Phase 4 → Human evaluation (Months 16-18) ▪ Collect and evaluate physiological data along with subjective surveys • Milestone → Quantitative/qualitative validation of the benefit of forest bathing			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	○ Lead stochastic modeling and simulation of forest lighting ○ Lead optimization and machine-learning framework development for parameter tuning ○ Lead development of statistical similarity and evaluation metrics ○ Lead human evaluation collection and data analysis in concert with biomedical professionals investigating correlations of visual pathways and mood regulation
Expected R&R of Korean SME Partner	○ Collaborating on research ideas and development/deployment of test lighting systems, along with human factors studies ○ Establish collaborations with top Korean vision/lighting/human factors experts familiar with the human visual system and mood regulation. ○ Assist with hardware development ○ Provide access to updated hardware platforms

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	○ Technological Outcomes ▪ Validated stochastic lighting model replicating natural forest environments ▪ Deployable, real-time indoor lighting control system ▪ Physiological evidence linking stochastic lighting to lowering stress ○ Commercial Potential ▪ Integrate the system into an office environment ▪ Combine with AR/VR for a much more immersive experience ○ Broader Impact ▪ Improved productivity and office space well-being ▪ Advance nature-inspired engineered indoor environments ▪ Personalized health-oriented building design

6. References

(1) B. J. Park, Y. Tsunetsugu, T. Kasetani, T. Kagawa, and Y. Miyazaki, “The physiological effects of Shinrin-yoku (taking in the forest atmosphere or forest bathing): evidence from field experiments in 24 forests across Japan,” Environ Health Prev Med, vol. 15, no. 1, pp. 18–26, Jan. 2010, doi: 10.1007/s12199-009-0086-9.

(2) J. A. Endler, “The Color of Light in Forests and Its Implications,” Ecological Monographs, vol. 63, no. 1, pp. 1–27, Feb. 1993, doi: 10.2307/2937121.

(3) L. E. Hartstein, A. Tuzikas, and R. F. Karlicek, “The Impact of Dynamic Changes in Light Spectral Power Distribution on Cognitive Performance and Wellbeing,” LEUKOS, vol. 16, no. 4, pp. 289–301, Oct. 2020, doi: 10.1080/15502724.2019.1693896.

(4) D. C. Fernandez et al., “Light Affects Mood and Learning through Distinct Retina-Brain Pathways,” Cell, vol. 175, no. 1, pp. 71-84. e18, Sep. 2018, doi: 10.1016/j.cell.2018.08.004.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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